

REDBUS NEO

Smart Bus Booking Platform

LISA Documentation

Student Name: Yatharth Dharmesh Mishra

Roll Number: 150096725055

Subject: React JS

Faculty/Guide: Prof. Prasad Junghare

College: ITM Skills University

Academic Year: 2025–2029

1. Project Overview

RedBus Neo is a modern bus booking platform developed using React.js. The application provides users with an intuitive interface to search for routes, compare buses, select seats, manage bookings, and view travel insights.

The project is inspired by real-world online bus reservation systems and demonstrates modern frontend development concepts including component-based architecture, state management, routing, responsive design, and local data persistence.

2. Problem Statement

Traditional bus ticket booking systems often suffer from:

- Limited route visibility
- Poor user experience
- Manual booking processes
- Difficulty comparing available buses
- Lack of centralized booking management

There is a need for a modern digital platform that simplifies bus booking while providing an enhanced user experience.

3. Objectives

The primary objectives of RedBus Neo are:

- Simplify the bus booking process
 - Provide an intuitive user interface
 - Enable efficient route discovery
 - Allow interactive seat selection
 - Manage bookings digitally
 - Demonstrate modern React.js development practices
 - Create a scalable and responsive frontend application
-

4. Technology Stack

Frontend Technologies

- React.js
- JavaScript (ES6+)
- Vite

Routing

- React Router

State Management

- React Context API

Styling

- CSS Modules

Data Persistence

- LocalStorage

Development Tools

- Visual Studio Code
 - Git
 - GitHub
-

5. Component Architecture

The application follows a modular component-based architecture.

User Interface



React Components



Context API State Management



Business Logic Layer



LocalStorage Persistence

Major Components

Layout Components

- Navbar
- Layout Wrapper

Search Components

- Search Form
- Bus Cards
- Filter Panel
- Sort Bar

Seat Selection Components

- Seat Grid
- Seat Cell
- Deck Switcher
- Price Summary

Booking Components

- Passenger Form
- Payment Module
- Booking Confirmation
- Cancellation Module

Dashboard Components

- Performance Dashboard
 - Achievements System
 - Travel Analytics
-

6. Key Features

Route Search

Users can search available buses using:

- Source city
 - Destination city
 - Travel date
-

Bus Filtering & Sorting

Available filters include:

- Price range
- AC / Non-AC
- Sleeper / Seater
- Departure time
- Ratings

Sorting options:

- Price
 - Rating
 - Departure Time
-

Interactive Seat Selection

Features:

- Visual seat layout
- Seat availability indicators

- Upper and lower deck support
 - Dynamic pricing
-

Passenger & Payment Module

Features:

- Passenger information entry
 - Form validation
 - Booking summary
 - Payment simulation
-

Booking Management

Users can:

- View upcoming bookings
 - View past bookings
 - Access booking history
 - Review ticket details
-

Ticket Cancellation

Features:

- Booking cancellation
 - Refund calculation
 - Cancellation tracking
 - Cancelled booking records
-

Advanced Features

Comfort Score

Provides a travel comfort rating for buses.

Crowd Meter

Displays estimated occupancy levels.

Fare Insights

Shows fare comparison information.

Carbon Savings

Displays environmental impact metrics.

Travel Conditions

Provides route-specific travel information.

Bus Personality Tags

Highlights unique bus characteristics.

Performance Dashboard

Provides:

- Booking statistics
 - User activity insights
 - Travel analytics
-

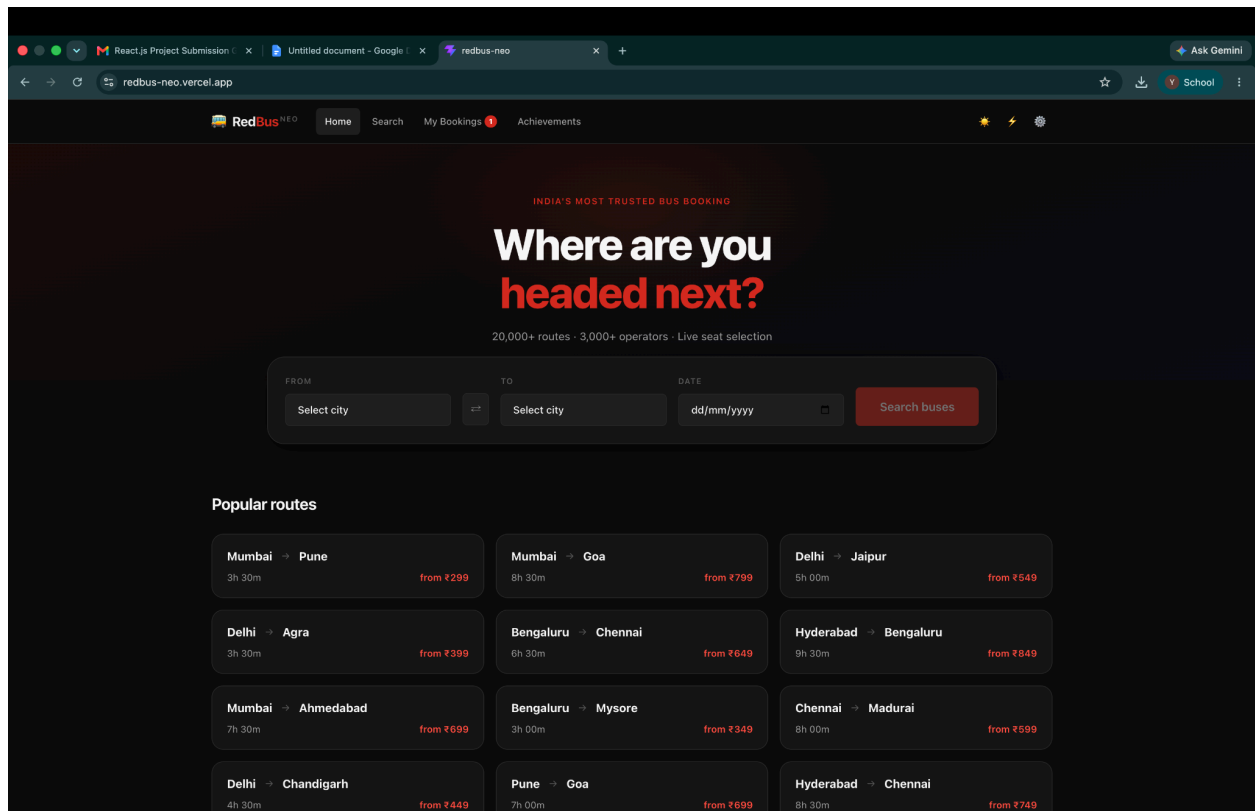
Achievements System

Gamification features include:

- Travel milestones
- Achievement badges
- User progress tracking

7. Screenshots

Home Page



Search Results

React.js Project Submission

Untitled document - Google

redbus-neo

redbus-neo.vercel.app/results?from=Mumbai&to=Pune&date=2026-06-15

Ask Gemini

RedBus

Home

Search

My Bookings

Achievements

FROM

TO

DATE

Mumbai

2nd

Pune

15/06/2026

Search

Mumbai → Pune

Mon, 15 Jun, 2026

Filters

AIR CONDITIONING

BUS TYPE

RATING

PRICE RANGE

AC

Non-AC

Sleeper

Seater

4.0+ rated only

MIN

MAX

₹ 199

₹ 520

5 buses found

Cheapest first

Fastest first

Highest rated

Earliest departure

N

National Travels

NON-AC SEATER

Budget Hero

07:30

MUMBAI

3h 22m

10:52

PUNE

★ 4.2

6.0

Moderate Demand

19 seats left

₹199

Select seats

S

SRS Travels

AC SEATER

Night Rider

04:30

MUMBAI

3h 20m

07:50

PUNE

★ 3.9

7.2

High Demand

9 seats left

₹260

Select seats

K

KSRTC

AC SLEEPER

Comfort King

20:30

MUMBAI

3h 35m

00:05

PUNE

★ 4.9

9.7

High Demand

9 seats left

₹310

Select seats

V

VRL Travels

LUXURY SLEEPER

Comfort King

17:30

MUMBAI

3h 13m

21:03

PUNE

★ 4.7

10.0

Low Demand

33 seats left

₹480

Select seats

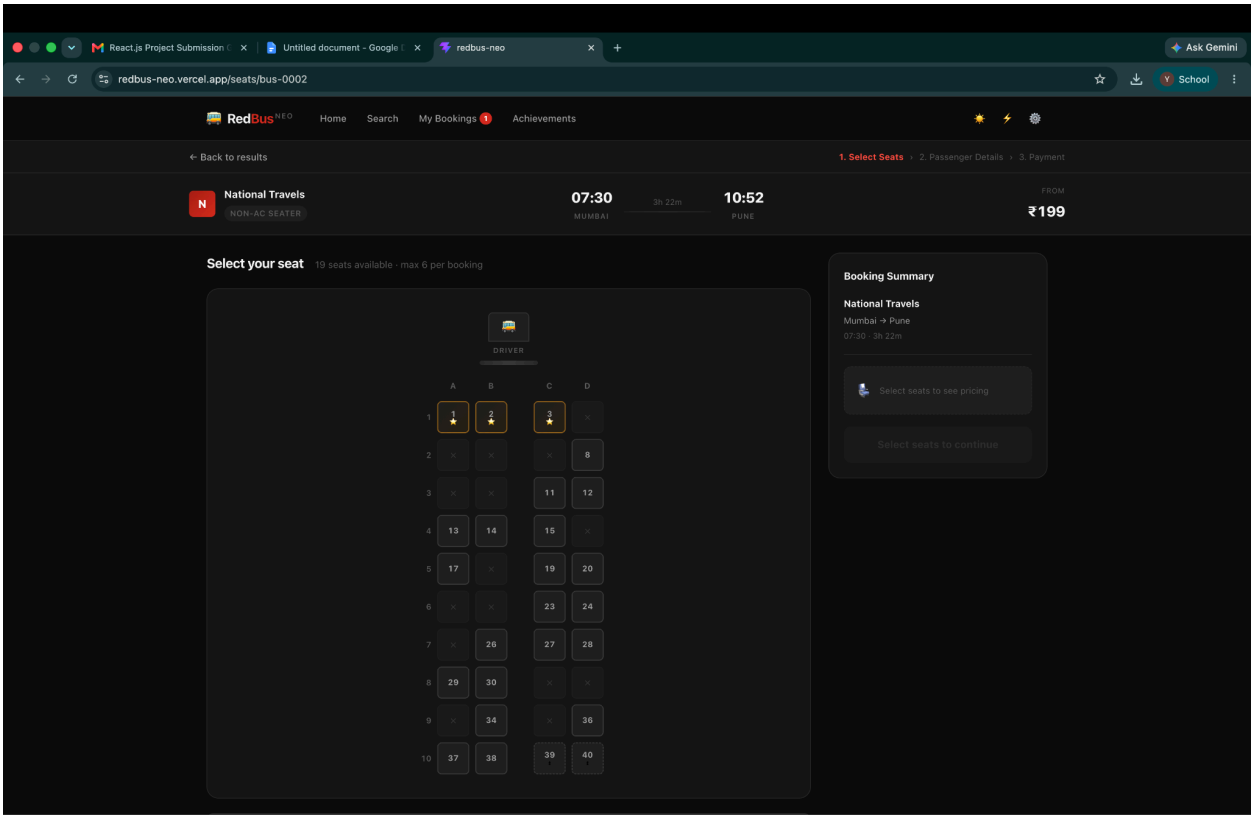
G

GSRTC

★ 4.4

FROM

Seat Selection



Passenger Details

React.js Project Submission

Untitled document - Google

redbus-neo

redbus-neo.vercel.app/passengers

Ask Gemini

RedBus

Home

Search

My Bookings

Achievements

1

2

3

4

Select Seats

Passengers

Payment

Confirmed

Passenger Details

Fill in details for each passenger

Passenger 1

Seat #8

FULL NAME *

e.g. Rahul Sharma

AGE *

25

GENDER *

Select

Order Summary

National Travels

Mumbai → Pune

07:30 · 3h 22m · Non-AC Seater

SEATS

Seat #8

₹209

Base fare

₹209

Service fee (5%)

₹10.00

Total payable

₹219

Inclusive of all taxes & fees.

Contact Details

Your ticket and booking updates will be sent here

MOBILE NUMBER *

+91 98765 43210

EMAIL ADDRESS *

you@example.com

JOURNEY DATE *

dd/mm/yyyy

Travel Add-ons

Optional services to enhance your journey

Travel Insurance

Covers cancellation, delay & medical emergencies

₹449

Extra Luggage (10kg)

Additional 10kg checked baggage allowance

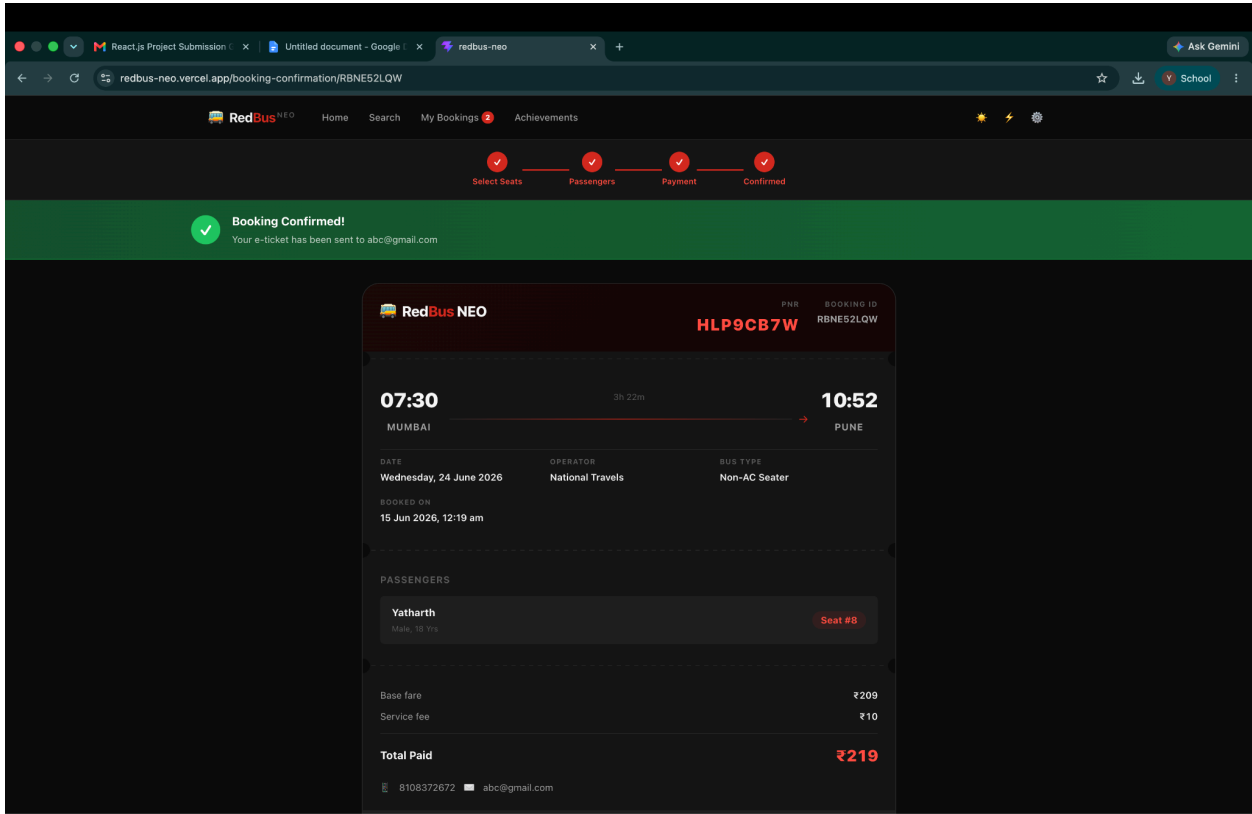
₹189

Priority Boarding

Board first, guaranteed overhead space

₹99

Booking Confirmation



My Bookings

React.js Project Submission

Untitled document - Google

redbus-neo

redbus-neo.vercel.app/my-bookings

Ask Gemini

RedBus

Home

Search

My Bookings

Achievements

My Bookings

2 bookings

Upcoming

PNR: HLP9CB7W

Confirmed

07:30

3h 22m

10:52

MUMBAI

PUNE

DATE

OPERATOR

SEATS

PASSENGERS

Wed, 24 Jun, 2026

National Travels

#8

1

₹219

Cancel Booking

View Ticket →

PNR: SSCEKC3K

Confirmed

07:30

0h 45m

14:15

PUNE

GOA

DATE

OPERATOR

SEATS

PASSENGERS

Mon, 22 Jun, 2026

Chartered Bus

#3

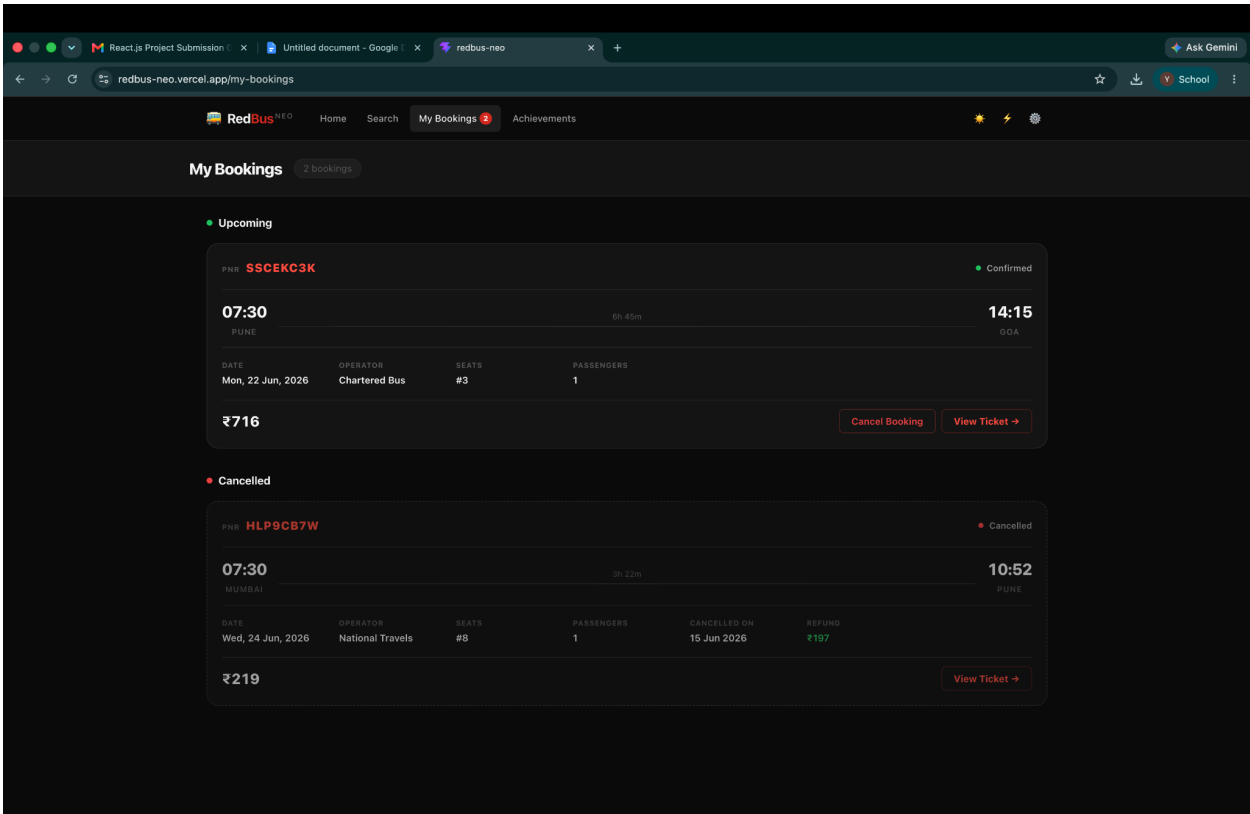
1

₹716

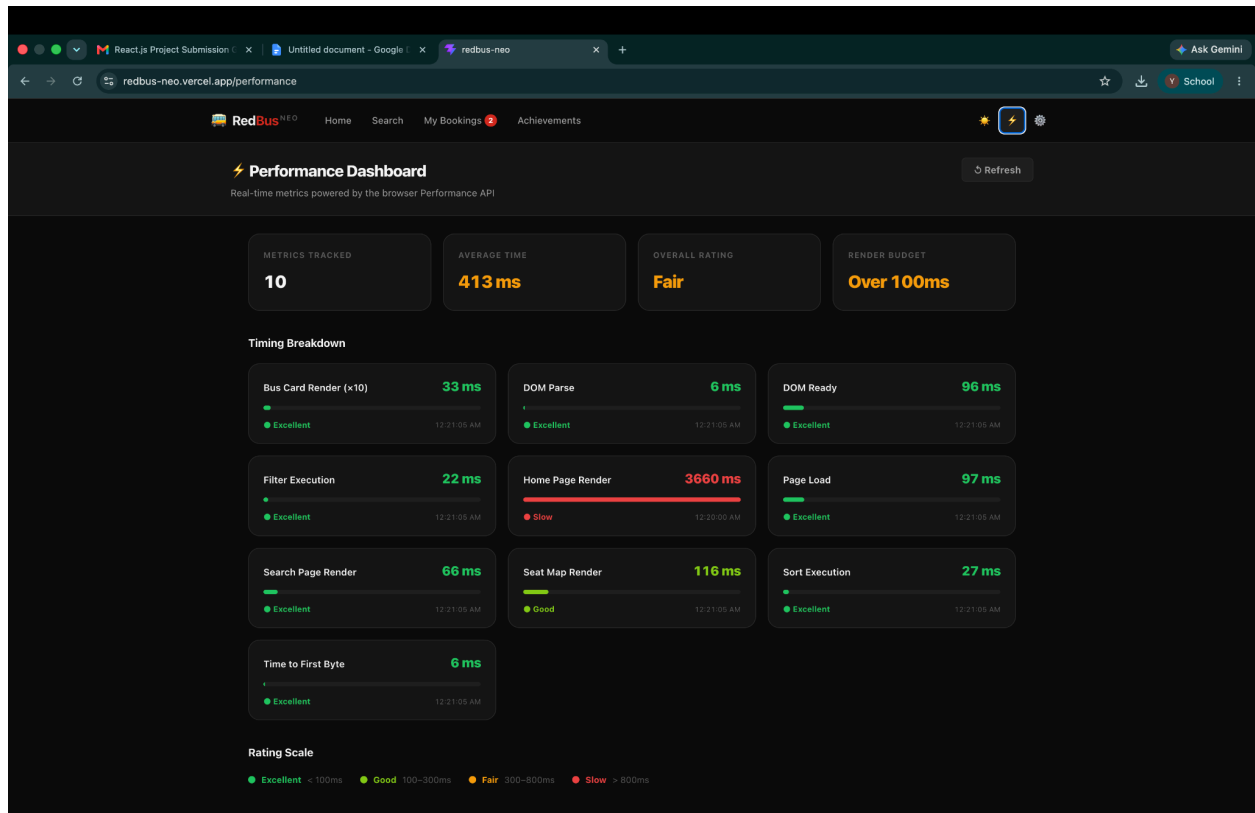
Cancel Booking

View Ticket →

Ticket Cancellation



Performance Dashboard



8. Project Outcomes

The project successfully achieved:

- Complete bus booking workflow
- Responsive user interface
- Modular React architecture
- Enhanced booking experience
- Expanded route and bus database
- Effective state management
- Local data persistence

9. Future Scope

Possible future enhancements include:

- Real-time GPS tracking
 - Live bus location monitoring
 - Backend integration
 - User authentication
 - Online payment gateway
 - Push notifications
 - AI-based route recommendations
-

10. Conclusion

RedBus Neo successfully demonstrates the implementation of a modern bus booking platform using React.js. The project incorporates routing, state management, responsive design, booking workflows, analytics, and user engagement features. It provides a practical example of frontend application development while delivering an intuitive and feature-rich user experience.

11. Project Links

GitHub Repository

<https://github.com/2025yatharthm-gif/redbus-neo>

Live Application

<https://redbus-neo.vercel.app/>